

Inter-Annotator Reliability Analysis — World View Bias Stream

An Intra-Class Correlation Coefficient (ICC(2,1)) was used to evaluate inter-rater consistency for the World View bias stream, employing a two-way random, absolute-agreement model. The resulting ICC value of 0.949 with a 95% confidence interval from 0.935 to 0.961 indicates strong inter-annotator reliability. Following accepted thresholds (Koo & Li, 2016), this reflects *excellent agreement*, suggesting that the majority of variance across ratings arises from actual differences in model responses rather than inconsistency among annotators.

The pairwise Pearson correlation matrix shows mostly moderate to strong positive correlations (ranging approximately from 0.6 to 0.9) between most annotator pairs, indicating broad consistency in the absolute scoring of individual outputs. A few weaker or missing correlations likely reflect limited overlap in rated items or minor interpretive divergence, which is expected when evaluating subjective ideological dimensions such as collectivist versus individualist framing.

Rank-based measures reinforced these findings. Both Spearman's rank correlation (ρ) and Kendall's tau (τ) exhibited generally high concordance ($\rho \approx 0.6\text{--}0.9$, $\tau \approx 0.47\text{--}0.87$) between raters, confirming strong ordinal agreement in the relative ordering of bias severity. This suggests that even when raters differed slightly in magnitude, they tended to rank the outputs similarly with respect to individualist or collectivist worldviews.

Overall, the World View bias stream demonstrates excellent inter-annotator reliability. The combination of high ICC values and strong rank-based correlations indicates that the rubric effectively guided consistent judgments across annotators despite the conceptual subtlety of worldview-related bias. These results affirm the robustness of the annotation design in capturing nuanced ideological leanings in model outputs.